

Computational Security and Pseudorandom Generators

CS 171: Cryptography

Sanjam Garg

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Overview

- 1 Computationally Secure Encryption
- 2 Pseudorandom Generators
- 3 Pseudo One-Time Pad
- 4 Security Proof of the Pseudo OTP
- 5 Practice: Manipulating PRGs
- 6 Summary

Defining Computationally Secure Encryption (Syntax)

Definition 1 (Private-Key Encryption Scheme)

A **private-key encryption scheme** is a tuple of algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$:

- $\text{Gen}(1^n)$: outputs a key k (we assume $|k| > n$).
- $\text{Enc}_k(m)$: takes key k and message $m \in \{0, 1\}^*$ as input; outputs ciphertext $c \leftarrow \text{Enc}_k(m)$.
- $\text{Dec}_k(c)$: takes key k and ciphertext c as input; outputs $m := \text{Dec}_k(c)$ or “error”.

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Correctness

For all n , all k output by $\text{Gen}(1^n)$, and all $m \in \{0, 1\}^*$:

$$\text{Dec}_k(\text{Enc}_k(m)) = m.$$

Computational Indistinguishability

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$

- 1 $\mathcal{A}$ outputs $m_0, m_1 \in \{0, 1\}^*$ with $|m_0| = |m_1|$.
- 2 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$, $c \leftarrow \text{Enc}_k(m_b)$ is given to $\mathcal{A}$.
- 3 $\mathcal{A}$ outputs a guess b' .
- 4 Output 1 if $b = b'$, and 0 otherwise.

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Definition 2 (Indistinguishable Encryptions (eavesdropper))

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ has **indistinguishable encryptions in the presence of an eavesdropper** if for all PPT $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Computational Indistinguishability: Two Relaxations

- **Perfect** indistinguishability requires $\Pr[\dots] \leq \frac{1}{2}$ for *all* adversaries $\mathcal{A}$.
- **Computational** indistinguishability relaxes this in two ways:
 - Quantify only over **PPT** (efficient) adversaries $\mathcal{A}$.
 - Allow a **negligible** advantage: $\frac{1}{2} + \text{negl}(n)$.

Remark

This definition **does not hide the message length!**

Constructing Secure Encryption

A building block

To construct computationally secure encryption we will use
pseudorandom generators (PRGs).

What Does It Mean To Be Random?

Is this string random?

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Definition 3 (Distribution / Uniform distribution)

A distribution on n -bit strings is a function $D : \{0, 1\}^n \rightarrow [0, 1]$ such that $\sum_x D(x) = 1$. For the **uniform** distribution on n -bit strings, denoted U_n , we set

$$\forall x \in \{0, 1\}^n : D(x) = 2^{-n}.$$

What About Pseudorandomness?

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- As before: **pseudorandomness is a property of a distribution**, and not of a specific string.

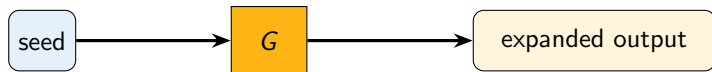
Pseudorandom Generators

- Stretches a short uniform “seed” into a larger “uniform-looking” output.

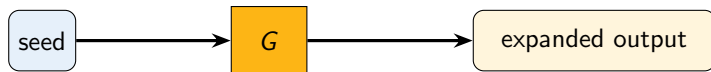
Pseudorandom Generators

- Stretches a short uniform “seed” into a larger “uniform-looking” output.
- Useful when only a few random bits are available.

Pseudorandom Generators (Formally)



Pseudorandom Generators (Formally)



Definition 4 (Pseudorandom Generator)

Let $\ell(\cdot)$ be a polynomial with $\ell(n) > n$. A deterministic poly-time algorithm $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ is a **pseudorandom generator** if for all PPT distinguishers $\mathcal{A}$ there is a negligible function $\text{negl}(\cdot)$ such that

$$\left| \Pr_{x \leftarrow U_{\ell(n)}} [\mathcal{A}(x) = 1] - \Pr_{s \leftarrow U_n} [\mathcal{A}(G(s)) = 1] \right| \leq \text{negl}(n).$$

- $\ell(\cdot)$ is the **stretch** of the generator.

PRG (Predicting-Game Style)

Game $\text{PRG}_{\mathcal{A},G}(1^n)$

- 1 $b \leftarrow \{0, 1\}$.
- 2 If $b = 0$ set $x \leftarrow G(U_n)$; else set $x \leftarrow U_{\ell(n)}$.
- 3 Give x to $\mathcal{A}$, who outputs a guess b' .
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Definition 5 (PRG, game form)

G is a **PRG** if for all PPT $\mathcal{A}$:

$$\Pr[\text{PRG}_{\mathcal{A},G}(1^n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Remark

The seed must be kept **secret** — analogous to the secret key in an encryption scheme.

$\text{PRG}_{\mathcal{A},G}$: Challenger vs. Adversary

$\text{PRG}_{\mathcal{A},G}(1^n)$



Definition 6 (PRG, game form)

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PRG_{A,G}: Challenger vs. Adversary

PRG_{A,G}(1ⁿ)

$b \leftarrow \{0, 1\}$
if $b = 0$: $x \leftarrow G(U_n)$ (pseudorandom)
if $b = 1$: $x \leftarrow U_{\ell(n)}$ (uniform)



Challenger



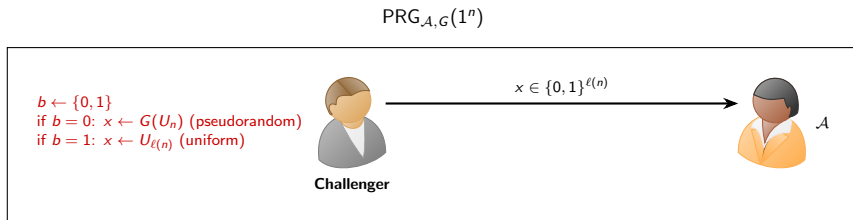
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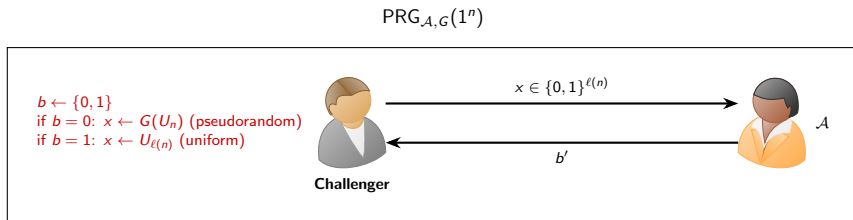


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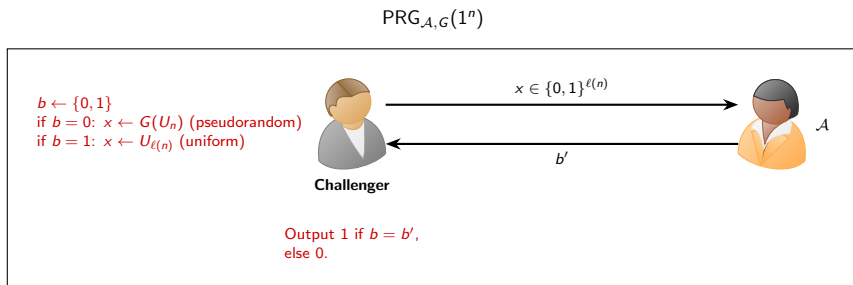


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Fixed-Length Encryption Scheme

Construction (Pseudo OTP)

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ be a PRG.

- $\text{Gen}(1^n)$: choose uniform $k \in \{0, 1\}^n$ and output it as the key.
- $\text{Enc}_k(m)$: on input a message $m \in \{0, 1\}^{\ell(n)}$, output the ciphertext

$$c := G(k) \oplus m.$$

- $\text{Dec}_k(c)$: on input a ciphertext $c \in \{0, 1\}^{\ell(n)}$, output the message

$$m := G(k) \oplus c.$$

- Like the one-time pad, but the pad is $G(k)$ instead of a truly random key.

Theorem 7

*If G is a PRG, then the construction above is a **fixed-length private-key encryption scheme** that has **indistinguishable encryptions** in the presence of an eavesdropper.*

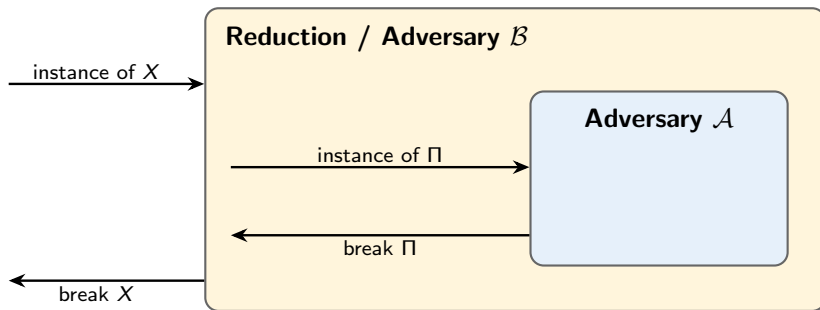
Proof by Reduction (If X then Π)

Strategy: if X is hard, then Π is secure

To prove: *if no PPT adversary breaks X (here, the PRG G), then no PPT adversary breaks Π .*

- Assume there exists a PPT $\mathcal{A}$ that “breaks” Π .
- Then we construct a PPT $\mathcal{B}$ that “breaks” X .
- But such a $\mathcal{B}$ cannot exist.
- Thus the assumption that some $\mathcal{A}$ breaks Π must have been false. □

Proof by Reduction: The Picture



Important

- **View of $\mathcal{A}$:** no change (it is run exactly as in its own game).
- $\mathcal{B}$ is PPT, given $\mathcal{A}$ is PPT.
- $\mathcal{B}$'s success degrades w.r.t. $\mathcal{A}$'s by at most $1/\text{poly}(n)$.

Theorem 8

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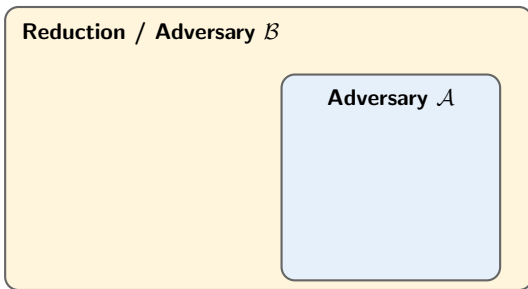
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Proof by reduction

Given a PPT adversary $\mathcal{A}$ “breaking” the encryption scheme, construct a PPT adversary $\mathcal{B}$ “breaking” the PRG.

Proof by Reduction (If PRG then Indistinguishable Encryption)

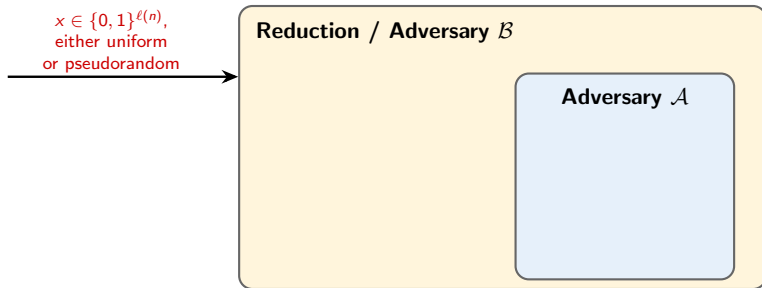


Given / To prove

Given: $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$.

To prove: $\Pr[\text{PRG}_{\mathcal{B}, G}(1^n) = 1] \geq \frac{1}{2} + \delta(n)$.

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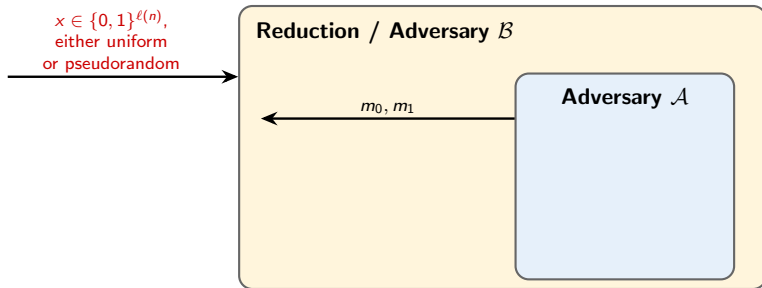


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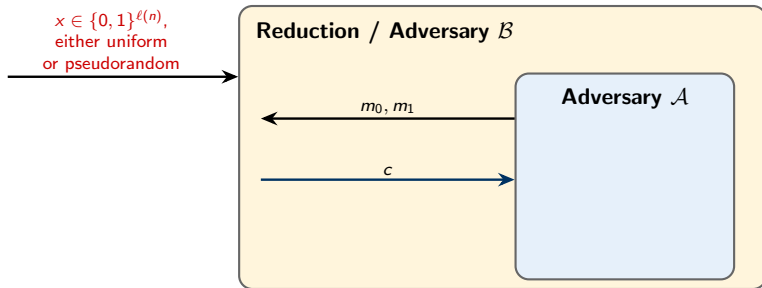


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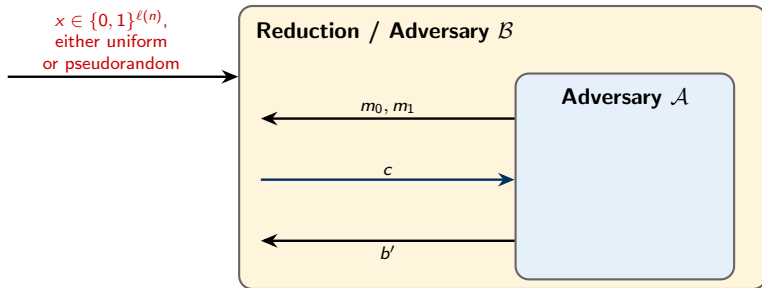


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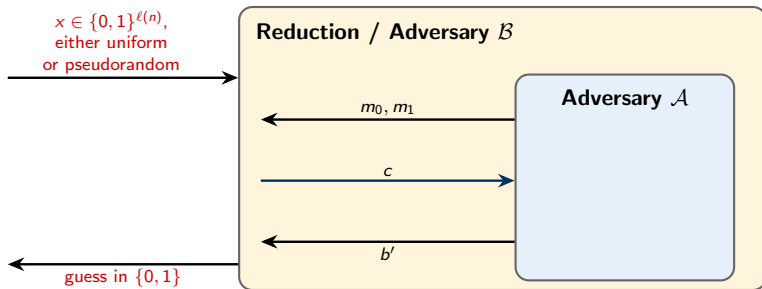


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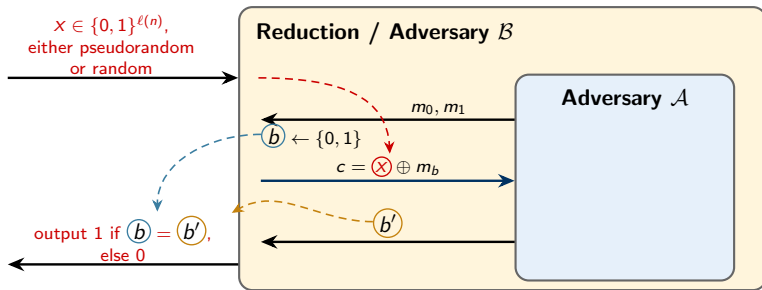


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Proof of Security: The Reduction $\mathcal{B}$



Reduction $\mathcal{B}$

- 1 Receive x .
- 2 Run $\mathcal{A}$; receive m_0, m_1 .
- 3 Sample $b \leftarrow \{0, 1\}$, set $c = x \oplus m_b$; give c to $\mathcal{A}$.
- 4 Receive b' . Output 1 ("pseudorandom") if $b = b'$, else 0 ("uniform").

Proof of Security: Two Cases

Case $x \leftarrow U_{\ell(n)}$ (uniform)

Like a one-time pad, so $\mathcal{A}$ has no advantage:
 $\Pr[b = b'] = \frac{1}{2}.$

Case $x \leftarrow G(U_n)$ (pseudorandom)

$\mathcal{A}$'s view is the real scheme, so
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If $x = G(k)$ for a uniform seed k , then $c = G(k) \oplus m_b$ — exactly the Pseudo OTP. So the view of $\mathcal{A}$ is exactly as in $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$.

Proof of Security: Putting It Together

Success probability of $\mathcal{B}$

$$\begin{aligned}\Pr[\mathcal{B} \text{ guesses correct}] &= \frac{1}{2} \Pr[\mathcal{B} \text{ correct} \mid x \leftarrow U_{\ell(n)}] + \frac{1}{2} \Pr[\mathcal{B} \text{ correct} \mid x \leftarrow G(U_n)] \\ &= \frac{1}{2} \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{1}{2} + \epsilon(n) \right) = \frac{1}{2} + \frac{\epsilon(n)}{2}.\end{aligned}$$

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Remark (running time)

$\mathcal{B}$ runs $\mathcal{A}$ once, performs one XOR, and a comparison — so $\mathcal{B}$ is PPT given $\mathcal{A}$ is PPT.

- **Pseudo OTP is secure** — assuming G is a PRG, with respect to our definition.
- **Gain**: Pseudo OTP has a **short key** — n bits instead of $\ell(n)$ bits.

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- **Gain**: Pseudo OTP has a **short key** — n bits instead of $\ell(n)$ bits.
- Does the Pseudo OTP allow encryption of *multiple* messages?
 - We will need to **define** it first!

Practice Question

Question

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a PRG. Is $H : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$, defined by $H(s) = (s \parallel 0^n) \oplus G(s)$, also a PRG — for *any* G ? Yes? No?

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Answer: No! — a counterexample

Let $F : \{0, 1\}^{n/2} \rightarrow \{0, 1\}^{3n/2}$ be a PRG and define, with $s_0, s_1 \in \{0, 1\}^{n/2}$,

$$G(s = (s_0, s_1)) = s_0 \parallel F(s_1).$$

- **Step 1:** prove G is a PRG.
- **Step 2:** prove H is *not* a PRG (for this G).

Step 1: G is a PRG — Proof by Reduction

Reduction / Adversary $\mathcal{B}$ (against F)

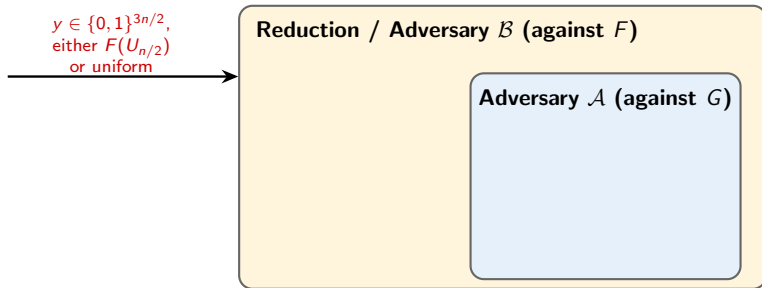
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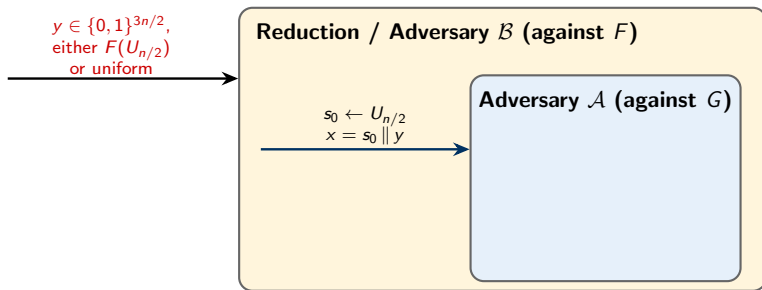


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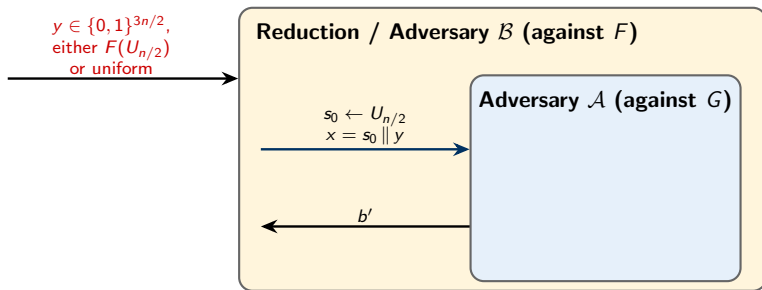


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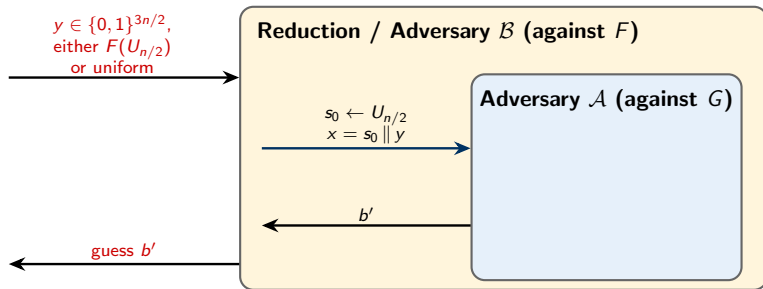


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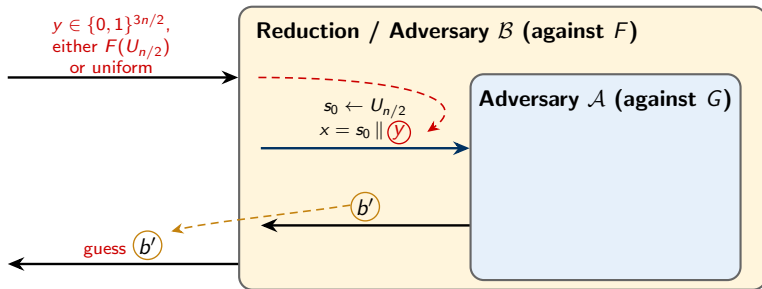


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Step 1: The Reduction $\mathcal{B}$



Reduction $\mathcal{B}$

- 1 Receive y .
- 2 Sample $s_0 \leftarrow U_{n/2}$; set $x = s_0 \parallel y \in \{0, 1\}^{2n}$.
- 3 Run $\mathcal{A}$ on x ; receive b' .
- 4 Output b' .

Step 1: Two Cases

Case $y \leftarrow U_{3n/2}$ (uniform)

Then $x = s_0 \parallel y$ is uniform in $\{0, 1\}^{2n}$ — exactly the $b = 1$ case of $\text{PRG}_{\mathcal{A}, G}$.

Case $y \leftarrow F(U_{n/2})$ (pseudorandom)

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Putting it together

In both cases $\mathcal{A}$'s view is exactly $\text{PRG}_{\mathcal{A}, G}(1^n)$ with the *same* challenge bit b , and $\mathcal{B}$ outputs $\mathcal{A}$'s guess. So

$$\Pr[\text{PRG}_{\mathcal{B}, F}(1^n) = 1] = \Pr[\text{PRG}_{\mathcal{A}, G}(1^n) = 1] \geq \frac{1}{2} + \epsilon(n),$$

contradicting that F is a PRG. Thus G is a PRG. □

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In both cases $\mathcal{A}$'s view is exactly $\text{PRG}_{\mathcal{A}, G}(1^n)$ with the *same* challenge bit b , and $\mathcal{B}$ outputs $\mathcal{A}$'s guess. So

$$\Pr[\text{PRG}_{\mathcal{B}, F}(1^n) = 1] = \Pr[\text{PRG}_{\mathcal{A}, G}(1^n) = 1] \geq \frac{1}{2} + \epsilon(n),$$

contradicting that F is a PRG. Thus G is a PRG. □

Remark (running time)

$\mathcal{B}$ samples $n/2$ bits, concatenates, and runs $\mathcal{A}$ once — so $\mathcal{B}$ is PPT given $\mathcal{A}$ is PPT.

Step 2: H is not a PRG

Computation

For this G we have $G(s) = s_0 \parallel F(s_1)$, so (writing $s = (s_0, s_1)$)

$$\begin{aligned} H(s) &= (s \parallel 0^n) \oplus G(s) = ((s_0 \parallel s_1) \parallel 0^n) \oplus (s_0 \parallel F(s_1)) \\ &= 0^{n/2} \parallel ((s_1 \parallel 0^n) \oplus F(s_1)). \end{aligned}$$

- The first $n/2$ output bits of H are always $0^{n/2}$.
- A distinguisher that outputs 1 iff the first $n/2$ bits are zero succeeds with overwhelming advantage.
- Hence H is **not** a PRG.



Summary

- **Computational security:** relax perfect secrecy to PPT adversaries with negligible advantage ($\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$).
- **Pseudorandomness** is a property of a *distribution*: indistinguishable from uniform by all PPT distinguishers.
- **PRG** $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ stretches a short secret seed into a longer pseudorandom string.
- **Pseudo OTP** $\text{Enc}_k(m) = G(k) \oplus m$ has indistinguishable encryptions if G is a PRG — proved by **reduction**.
- **Gain:** a short (n -bit) key encrypts an $\ell(n)$ -bit message.
- PRGs do not compose naively: $H(s) = (s \parallel 0^n) \oplus G(s)$ need not be a PRG.